

Bernard Lee

DATA TECHNOLOGY & AI LEADER

Hong Kong +852 5243 7393 b@nggdim.com github.com/pybern linkedin.com/in/bernard-lee-57042b50

Technology leader spanning enterprise data architecture, AI, analytics, and digital delivery. Combines hands-on engineering with architecture governance, business-case development, and cross-functional leadership—translating complex business needs into secure, scalable platforms with measurable operational and commercial outcomes.

EXPERIENCE HIGHLIGHTS

Hong Kong Monetary Authority Technology Innovation Lead

Jul 2025–Present

- Re-architected a four-service Azure data-fetching solution into one containerized AWS application within a month, preserving performance across four sources while improving usability and error visibility.
- Built a natural-language analytics PoC in three months for five pilot users, orchestrating schema discovery, RAG, SQL generation, validation, execution, and AI visualization over permission-controlled warehouse data.
- Developed a multi-agent risk-profiling PoC that processed up to thousands of external articles per run and attached confidence scores and evidence to semantic risk tags.
- Led the Architecture Review Board-approved design of an enterprise API and AI gateway for up to 100M transactions, covering semantic caching, observability, routing, fallback, and model controls.

Hong Kong Monetary Authority Data Architect

Nov 2023–Jun 2025

- Reviewed data flows, architecture diagrams, and proposals for 10–15 projects, providing governance recommendations on secure and scalable data design.
- Defined the organization's approved medallion reference architecture and reusable patterns for ingestion, transformation, storage, and consumption.
- Established secure web-scraping patterns adopted by at least five teams and built concurrent Python ingestion pipelines that cut retrieval from five hours to under one.
- Delivered architecture and AI knowledge-sharing sessions for audiences ranging from 25 to more than 300 people.

MCM Worldwide

Jul 2022–Oct 2023

Principal Data Architect

- Delivered Global Finance and CEO dashboards across five countries, supported by pipelines spanning the enterprise SQL Server data warehouse.
- Created optimized warehouse views that improved data accuracy and reduced Power BI query response from minutes to seconds.
- Introduced modular patterns across 20–30 dashboard and query-performance components and led two BI specialists.

AIA Hong Kong & Macau

Jan 2019–Jul 2022

Associate Director · Senior Manager, Health & Wellness

- Combined claims, health-program, engagement, and participant data for weekly reporting and quarterly wellness-planning analysis across three clients and 6,000+ employees.
- Increased program adoption by 25 percentage points; analysis associated participation with 5–10% lower claims costs.
- Productionized 90%-accurate Azure OCR for scoped claims, replacing 25% of manual QA workload, and developed NLP-based keyword categorization for health data.
- Built business cases covering US\$3M in analytics capability, US\$1M in application development, and multiple PoCs.

PROFILE

Sheffield Hallam University

BEng Electrical & Electronics Engineering
First City University College · 2010

Hong Kong Permanent Resident
English · Cantonese

Earlier data experience

KPMG Malaysia & US · 2016–2018

- Built a Python NLP framework for approximately 25M financial communications; reuse reduced a later investigation's delivery time by 67%.
- Analyzed approximately 5M transactions plus SWIFT and CHIPS messages during a US secondment.

AIA Bhd. · 2012–2016

- Led the first cross-functional Tableau deployment for approximately 50 users and delivered 10+ reports.
- Built pipelines across four source systems and accelerated reporting by up to 10× with Python, SQL, and VBA.

Core data stack

Python

SQL

Databricks

SQL Server

Power BI

Tableau

SAS

D3.js

AWS

Azure

Data Pipelines

Medallion

LLMs

RAG

NLP

OCR

Best-fit focus

Enterprise data architecture · Data platforms · Data governance · Data engineering · AI & analytics · Financial and insurance data